

since we think of the starting point $\mathbf{q}_0$ as fixed. To calculate the derivative of $\hat{\mathcal{A}}$ with respect to $\mathbf{q}_n$, $n = 1, 2, \dots, v-1$, note that only a few terms of the discrete action involve this configurational point, thus

$$\begin{aligned} \frac{\partial \hat{\mathcal{A}}}{\partial \mathbf{q}_n} &= \frac{\partial}{\partial \mathbf{q}_n} \left[\frac{1}{2h} ((\mathbf{q}_n - \mathbf{q}_{n-1})^T \mathbf{M}(\mathbf{q}_n - \mathbf{q}_{n-1}) + (\mathbf{q}_{n+1} - \mathbf{q}_n)^T \right. \\ &\quad \times \left. \mathbf{M}(\mathbf{q}_{n+1} - \mathbf{q}_n)) - U(\mathbf{q}_n)h \right] \\ &= \frac{1}{h} [\mathbf{M}(\mathbf{q}_n - \mathbf{q}_{n-1}) - \mathbf{M}(\mathbf{q}_{n+1} - \mathbf{q}_n)] - \nabla U(\mathbf{q}_n)h, \end{aligned}$$

which yields the equations

$$\mathbf{M}(\mathbf{q}_{n+1} - 2\mathbf{q}_n + \mathbf{q}_{n-1}) = -h^2 \nabla U(\mathbf{q}_n), \quad n = 1, 2, \dots, v-1. \quad (2.4)$$

We can think of

$$\frac{1}{h^2} (\mathbf{q}_{n+1} - 2\mathbf{q}_n + \mathbf{q}_{n-1})$$

as a centered finite difference approximation of the acceleration (the second derivative of position), thus the Eq. (2.4) is a direct discretization of Newton's equations.

Somewhat surprisingly, the one-sided approximation of velocities in the discrete Lagrangian has led to a symmetric discretization of the equations of motion. There is a slight issue of what to do at the endpoints. In the variational formulation we think of the endpoints of the curve as fixed points which means that we are effectively solving a boundary value problem and Eq. (2.4) tells us how to compute all the interior points along the path. In the case of an initial value problem (the usual issue in molecular dynamics) we do not know in advance the right endpoint value, but we assume that we can, in some way, calculate $\mathbf{q}_1$ (by a *starting procedure*) and so Eq. (2.4) defines $\mathbf{q}_2, \mathbf{q}_3, \dots, \mathbf{q}_v$.

The method (2.4) is commonly referred to as *Störmer's rule*. It was used by the mathematician Störmer for calculations in the first decade of the 1900s. In molecular dynamics this method is referred to as the Verlet method since it was used by Verlet in his important 1967 paper [387].

The scheme is usually given in an alternative “velocity Verlet” form that takes a step from a given vector $\mathbf{q}_n, \mathbf{v}_n$ to $\mathbf{q}_{n+1}, \mathbf{v}_{n+1}$ by the sequence of operations

$$\mathbf{v}_{n+1/2} = \mathbf{v}_n + (h/2)\mathbf{M}^{-1}\mathbf{F}_n, \quad (2.5)$$

$$\mathbf{q}_{n+1} = \mathbf{q}_n + h\mathbf{v}_{n+1/2}, \quad (2.6)$$

$$\mathbf{v}_{n+1} = \mathbf{v}_{n+1/2} + (h/2)\mathbf{M}^{-1}\mathbf{F}_{n+1}, \quad (2.7)$$

$$\tilde{U}(\mathbf{q}) = U(\mathbf{q}) - \frac{h^2}{24} \nabla U(\mathbf{q})^T \mathbf{M}^{-1} \nabla U(\mathbf{q}).$$

The forces arising from such a modified potential can be worked out:

$$\tilde{\mathbf{F}} = -\nabla \tilde{U} = -\left[\mathbf{I} + \frac{h^2}{12} \mathbf{U}'' \mathbf{M}^{-1} \right] \nabla U,$$

where $\mathbf{U}''$ is the Hessian matrix of the potential. It may in some cases be a daunting task to compute the Hessian matrix as would be required in the Takahashi-Imada method, but this is not always the case. Advantage can be made of the fact that for systems with short-ranged potentials, the Hessian matrix is likely to be very sparse, in which case this computation can be effected with little additional overhead, although this will depend on the underlying computer architecture, the size of the system, etc.

This method can be shown to have effective order four, meaning that there is a change of variables χ_h which can be used to transform the Takahashi-Imada method into one of order four using the processing technique of Sect. 2.4.5. The potential energy modification has been specifically chosen to annihilate terms in the local error expansion (after coordinate transformation).

A discussion and comparison of several multiderivative methods for molecular applications may be found in [239].

2.5.5 Other Methods

We have mentioned previously that it is possible to reduce the Verlet method to a scheme involving positions only:

$$\mathbf{q}_{n+1} - 2\mathbf{q}_n + \mathbf{q}_{n-1} = h^2 \mathbf{M}^{-1} \mathbf{F}(\mathbf{q}_n).$$

This is a type of a *multistep* method. Such methods may be studied using a generalization of the techniques used to understand one-step methods [167]. There are a variety of multistep methods which could in principle be used for molecular dynamics, however, we regard the benefits as unproven; in particular, such methods neglect the phase space structure such as the symplectic property.

Another scheme which is sometimes used in molecular dynamics is Beeman's Algorithm [30, 331].